

Physical Chemistry (I)

Lecture 23

Electrochemical Cells



In the Last Class

- Review: electrostatics
- Physical chemistry of electrolytes
 - Gibbs energy for solvation: Born equation
 - Potential distributions: Debye-Hückel equation
 - Activities: Debye-Hückel limiting law

Born Equation

$$\Delta_{\text{solv}} G^{\ominus} = -\frac{z_i^2 e^2}{8\pi\epsilon_0 r_i} \left(1 - \frac{1}{\epsilon_r}\right)$$

where the relative permittivity $\epsilon_r = \epsilon/\epsilon_0$.

- $\Delta_{\text{solv}} G^{\ominus} < 0$: solvation of ions is favored
- Water has $\epsilon_r = 78.54$ at 25°C, and

$$\Delta_{\text{solv}} G^{\ominus} = -\frac{z_i^2}{r_i/\text{pm}} \times (6.86 \times 10^4 \text{ kJ mol}^{-1})$$

Debye-Hückel Equation

- Poisson-Boltzmann equation

$$\nabla^2 \phi = \frac{zF n_{\infty}}{\varepsilon V} \left(e^{+ze\phi/k_B T} - e^{-ze\phi/k_B T} \right)$$

- Debye-Hückel equation

$$\nabla^2 \phi = \phi / r_D^2$$

where the Debye radius r_D is

$$r_D = \sqrt{\frac{\varepsilon R T}{2 F^2 I b^{\ominus} (m_A / V)}}$$

Ion Activities

- Chemical potential of ion

$$\mu_J = \mu_J^{\text{ideal}} + RT \ln \gamma_{\pm}$$

where the mean activity coefficient is $\gamma_{\pm} = (\gamma_+^p \gamma_-^q)^{1/(p+q)}$

- Molar work of charging an ion

$$w_{e,m} = -\frac{z_i^2 F^2}{8\pi N_A \epsilon r_D}$$

- Debye-Hückel limiting law

$$\log_{10} \gamma_{\pm} = -A |z_+ z_-| I^{1/2}$$

Several Types of Work

$$\delta w = (\text{intensive factor}) \times (\text{extensive factor})$$

Type of work	dw	Comments	Units [†]	
Expansion	$-p_{\text{ex}}dV$	p_{ex} is the external pressure dV is the change in volume	Pa m^3	} δw_{exp}
Surface expansion	$\gamma d\sigma$	γ is the surface tension $d\sigma$ is the change in area	N m^{-1} m^2	
Extension	$f dl$	f is the tension dl is the change in length	N m	} δw_{add}
Electrical	ϕdQ	ϕ is the electric potential dQ is the change in charge	V C	
	$Q d\phi$	$d\phi$ is the potential difference	V	
		Q is the charge transferred	C	



Last Generalization

For a non-expansion work $\delta w = f dX$,

- State: $(p, T, n_A, n_B, \dots) \rightarrow (p, T, X, n_A, n_B, \dots)$
- Thermodynamic potentials
 - $dU = TdS - pdV + f dX + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dH = TdS + Vdp + f dX + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dA = -SdT - pdV + f dX + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dG = -SdT + Vdp + f dX + \mu_A dn_A + \mu_B dn_B + \dots$

Examples

- Surface tension

$$dU = TdS - pdV + \gamma d\sigma + \mu_A dn_A + \mu_B dn_B + \dots$$

- Electric work (Q as an independent variable)

$$dU = TdS - pdV + \phi dQ + \mu_A dn_A + \mu_B dn_B + \dots$$

- Electric work (ϕ as an independent variable)

$$dU = TdS - pdV + Q d\phi + \mu_A dn_A + \mu_B dn_B + \dots$$

Not in Atkins

Electrochemical Reactions

For an electrolytic solution,

$$dU = TdS - pdV + \phi dQ + \sum_J \mu_J dn_J$$

- Ions: charge carriers

$$Q = \sum_J z_J F n_J$$

$$dQ = \sum_J z_J F dn_J$$

Electrochemical Potentials

$$dU = TdS - pdV + \phi \sum_J z_J F dn_J + \sum_J \mu_J dn_J$$

$$dU = TdS - pdV + \sum_J (z_J F \phi + \mu_J) dn_J$$

Define the **electrochemical potential**:

$$\mu'_J = z_J F \phi + \mu_J$$

$$dU = TdS - pdV + \sum_J \mu'_J dn_J$$

“Diffusive” Equilibrium

$$dU = TdS - pdV + \sum_J \mu'_J dn_J$$

At (diffusive) equilibrium of the electrolytic solution,
 μ'_J is the same in and throughout every phase in the system.

Distribution of Ions

$$\mu'_J(x) = \mu_J(x) + z_J F \phi(x) = \mu_J^\ominus + RT \ln a_J + z_J F \phi(x)$$

Assuming that $a_J \approx [J]/c^\ominus = n_J/(c^\ominus V)$,

$$\mu'_J(x) = \mu_J^\ominus + RT \ln(n_J/c^\ominus V) + z_J F \phi(x)$$

Hence, the equilibrium condition $\mu'_J(x_1) = \mu'_J(x_2)$ becomes

$$RT \ln n_J(x_1) + z_J F \phi(x_1) = RT \ln n_J(x_2) + z_J F \phi(x_2)$$

or,

$$n_J(x_2) = n_J(x_1) \exp \left[-\frac{z_J F \{\phi(x_2) - \phi(x_1)\}}{RT} \right]$$

Electrode-Ion Equilibrium

- Equilibrium between the electrode (solid) and ions (solute)

$$\mu'_J(\text{solid}) = \mu'_J(\text{solute})$$

$$\mu'_J(\text{solid}) = \mu_J^\ominus(\text{solid}) + z_J F \phi_{\text{solid}}$$

$$\mu'_J(\text{solute}) = \mu_J^\ominus(\text{solute}) + RT \ln a_J + z_J F \phi_{\text{solute}}$$

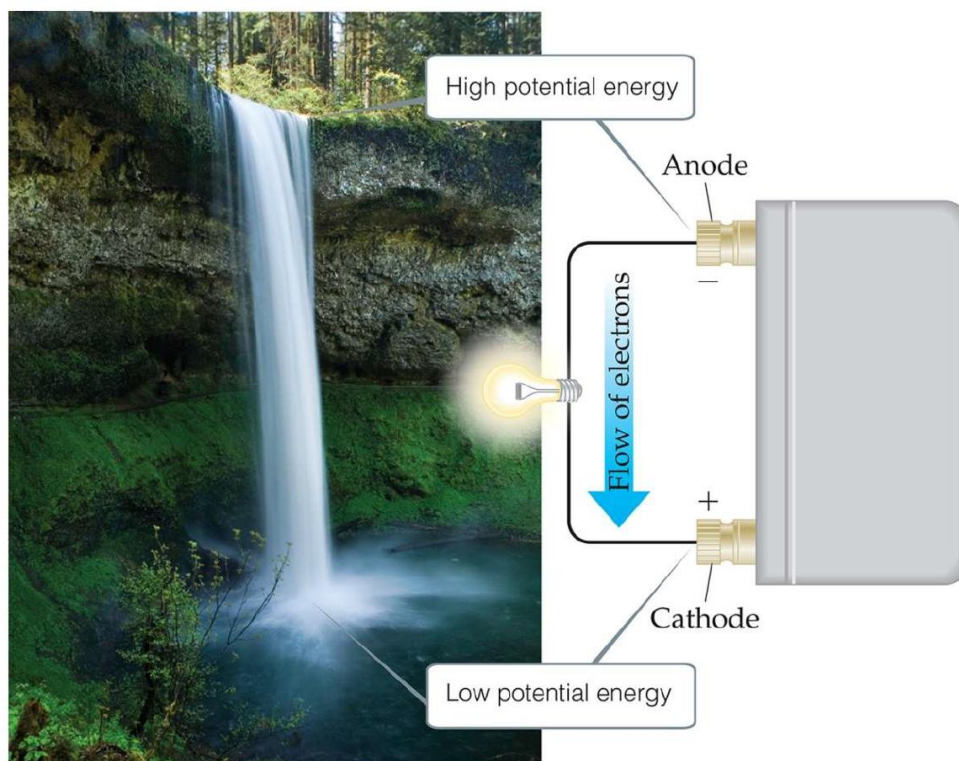
Hence,

$$\phi_{\text{solute}} = \phi_0 - \frac{RT}{z_J F} \ln a_J$$

where $\phi_0 = \{\mu_J^\ominus(\text{solid}) - \mu_J^\ominus(\text{solute})\} / z_J F + \phi_{\text{solid}}$

Cell Potential

Potential difference between the two electrodes



(Image: <https://www.unf.edu/~michael.lufaso/chem2046/2046chapter20.pdf>)

Thermodynamics of Cell

Recall that

$$dG_{T,p} = \delta w_{\text{add,max}} = \delta w_{\text{elec}}$$

for a **reversible** electrochemical reaction.

For an infinitesimal progress of the reaction $d\xi$,

$$dG = \Delta_r G d\xi$$

Hence,

$$\delta w_{\text{elec}} = \Delta_r G d\xi$$

Electrical Work

$$\delta w_{\text{elec}} = \phi dQ$$

where ϕ is the electric field and dQ is the transferred charge.

For the electrochemical reaction,

$$\phi = E_{\text{cell}}$$

$$dQ = -\nu e N_A d\xi = -\nu F d\xi$$

Hence,

$$\delta w_{\text{elec}} = -\nu F E_{\text{cell}} d\xi$$

Thermodynamics of Cell

$$\delta w_{\text{elec}} = \Delta_r G d\xi$$

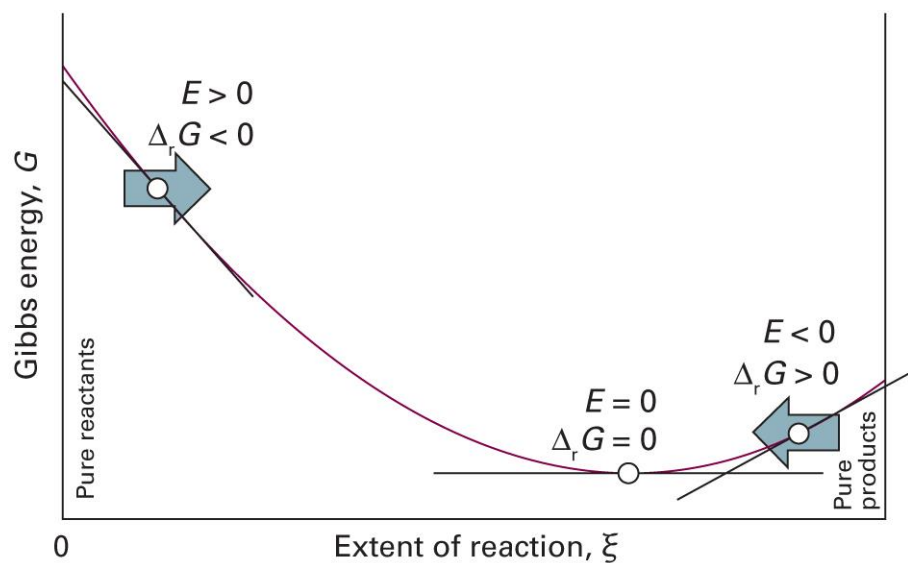
$$\delta w_{\text{elec}} = -\nu F E_{\text{cell}} d\xi$$

Combining the two, we obtain

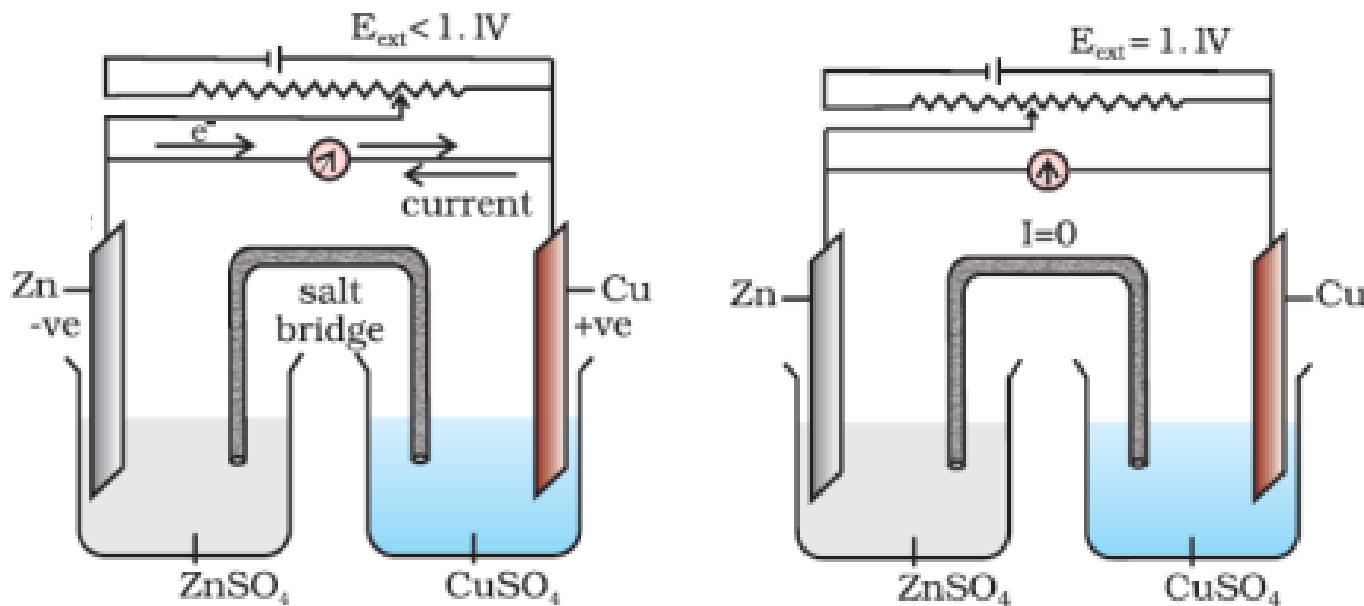
$$\Delta_r G = -\nu F E_{\text{cell}}$$

Cell Potential and Reaction

$$\Delta_r G = -\nu F E_{\text{cell}}$$



Measurement of Cell Potential



$$E_{\text{cell}} = (E_{\text{ext}})_{I=0}$$

Nernst Equation

$$\Delta_r G = -\nu F E_{\text{cell}}$$

Using $\Delta_r G = \Delta_r G^\ominus + RT \ln Q$,

$$-\nu F E_{\text{cell}} = \Delta_r G^\ominus + RT \ln Q$$

Thus,

$$E_{\text{cell}} = E_{\text{cell}}^\ominus - \frac{RT}{\nu F} \ln Q$$

where $E_{\text{cell}}^\ominus = -\Delta_r G^\ominus / \nu F$

At Equilibrium

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln Q$$

At equilibrium, $E_{\text{cell}} = 0$ and $Q = K$

$$0 = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln K$$

or,

$$E_{\text{cell}}^{\ominus} = \frac{RT}{\nu F} \ln K$$

Temperature Coefficient

$$E_{\text{cell}}^{\ominus} = -\frac{\Delta_{\text{r}}G^{\ominus}}{\nu F}$$

Since $(\partial G / \partial T)_p = -S$,

$$\frac{dE_{\text{cell}}^{\ominus}}{dT} = -\frac{1}{\nu F} \left(\frac{\partial \Delta_{\text{r}}G^{\ominus}}{\partial T} \right)_p = \frac{\Delta_{\text{r}}S^{\ominus}}{\nu F}$$

Furthermore,

$$\Delta_{\text{r}}H^{\ominus} = \Delta_{\text{r}}G^{\ominus} + T\Delta_{\text{r}}S^{\ominus} = -\nu F \left(E_{\text{cell}}^{\ominus} - T \frac{dE_{\text{cell}}^{\ominus}}{dT} \right)$$

Electrode Potentials

Cell potential = cathode contribution + anode contribution

For a cell L||R,

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{R}) - E^{\ominus}(\text{L})$$

Impossible to measure each contribution

Convention

- Standard hydrogen electrode (SHE): $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq})$

$E^\ominus = 0$ at all temperatures

- Standard potential $E^\ominus(\text{X})$

For a cell $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq}) || \text{X}$, $E^\ominus(\text{X}) = E_{\text{cell}}^\ominus$

Standard Potentials

Couple	$E^\ominus/\text{V}$
$\text{Ce}^{4+}(\text{aq}) + \text{e}^- \rightarrow \text{Ce}^{3+}(\text{aq})$	+1.61
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{AgCl}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq})$	+0.22
$\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2(\text{g})$	0
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71

Thermodynamics of Potentials

Each electrode potential can be converted into

$$-vFE^{\ominus} = \Delta_r G^{\ominus}$$

When a combination of reactions a and b gives reaction c,

$$\Delta_r G^{\ominus}(c) = \Delta_r G^{\ominus}(a) + \Delta_r G^{\ominus}(b)$$

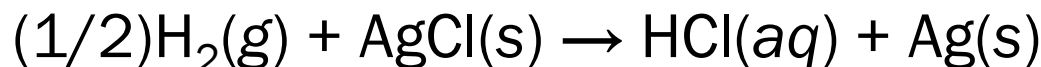
$$-v_c F E^{\ominus}(c) = -v_a F E^{\ominus}(a) - v_b F E^{\ominus}(b)$$

Thus,

$$v_c E^{\ominus}(c) = v_a E^{\ominus}(a) + v_b E^{\ominus}(b)$$

Harned Cell

Consider the Harned cell:



- Cell potential

$$E_{\text{cell}}^{\ominus} = E^{\ominus}(\text{AgCl/Ag, Cl}^{-}) - E^{\ominus}(\text{SHE}) = E^{\ominus}(\text{AgCl/Ag, Cl}^{-})$$

- Nernst equation

$$E_{\text{cell}} = E^{\ominus}(\text{AgCl/Ag, Cl}^{-}) - \frac{RT}{F} \ln \frac{a(\text{H}^{+})a(\text{Cl}^{-})}{\{a(\text{H}_2)\}^{1/2}}$$

Activities

$$E_{\text{cell}} = E^{\ominus}(\text{AgCl}/\text{Ag}, \text{Cl}^{-}) - \frac{RT}{F} \ln \frac{a(\text{H}^{+})a(\text{Cl}^{-})}{\{a(\text{H}_2)\}^{1/2}}$$

- Let $E^{\ominus} = E^{\ominus}(\text{AgCl}/\text{Ag}, \text{Cl}^{-})$
- At the standard pressure of 1 bar, $a(\text{H}_2) = 1$
- For the molality b of $\text{HCl}(\text{aq})$,

$$a(\text{H}^{+}) = \gamma_{\pm} b / b^{\ominus} \text{ and } a(\text{Cl}^{-}) = \gamma_{\pm} b / b^{\ominus}$$

$$E_{\text{cell}} = E^{\ominus} - \frac{RT}{F} \ln \left(\frac{\gamma_{\pm} b}{b^{\ominus}} \right)^2$$

Molality Dependence

$$E_{\text{cell}} = E^{\ominus} - \frac{RT}{F} \ln \left(\frac{\gamma_{\pm} b}{b^{\ominus}} \right)^2 = E^{\ominus} - \frac{2RT}{F} \left(\ln \gamma_{\pm} + \ln \frac{b}{b^{\ominus}} \right)$$

From the Debye-Hückel limiting law,

$$\begin{aligned} \ln \gamma_{\pm} &= (\ln 10) \log \gamma_{\pm} = -(\ln 10) A |z_+ z_-| I^{1/2} \\ &= -(A \ln 10) (b/b^{\ominus})^{1/2} \end{aligned}$$

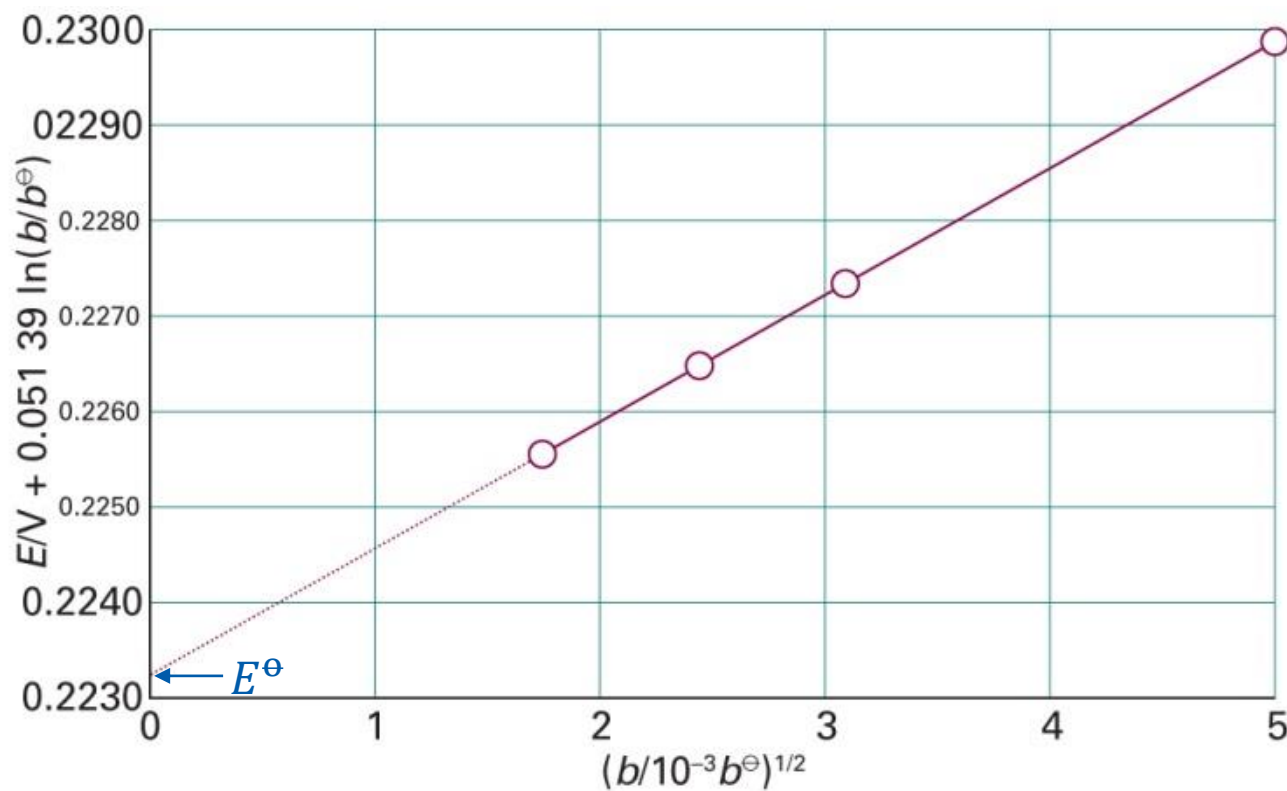
$$E_{\text{cell}} + \frac{2RT}{F} \ln \frac{b}{b^{\ominus}} = E^{\ominus} - \frac{2RT}{F} \ln \gamma_{\pm} = E^{\ominus} + \frac{2A R T \ln 10}{F} \sqrt{\frac{b}{b^{\ominus}}}$$

Determination of $E^\ominus$

$$\underbrace{E_{\text{cell}} + \frac{2RT}{F} \ln \frac{b}{b^\ominus}}_y = E^\ominus + \underbrace{\frac{2ART \ln 10}{F}}_a \underbrace{\sqrt{\frac{b}{b^\ominus}}}_x$$

From the plot of x versus y ,
we can obtain $E^\ominus$ as the y -intercept

Example



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 6D.1)

Determination of $\gamma_{\pm}$

$$E_{\text{cell}} + \frac{2RT}{F} \ln \frac{b}{b^{\ominus}} = E^{\ominus} - \frac{2RT}{F} \ln \gamma_{\pm}$$

From the measurement of E_{cell} , we can obtain

$$\ln \gamma_{\pm} = \frac{F}{2RT} (E^{\ominus} - E_{\text{cell}}) - \ln \frac{b}{b^{\ominus}}$$

Summary

